

Coding Agent Memory Post-training

Unlocking the Memory Potential of Pre-trained File Operations for Long-Horizon Tasks via Reinforcement Learning

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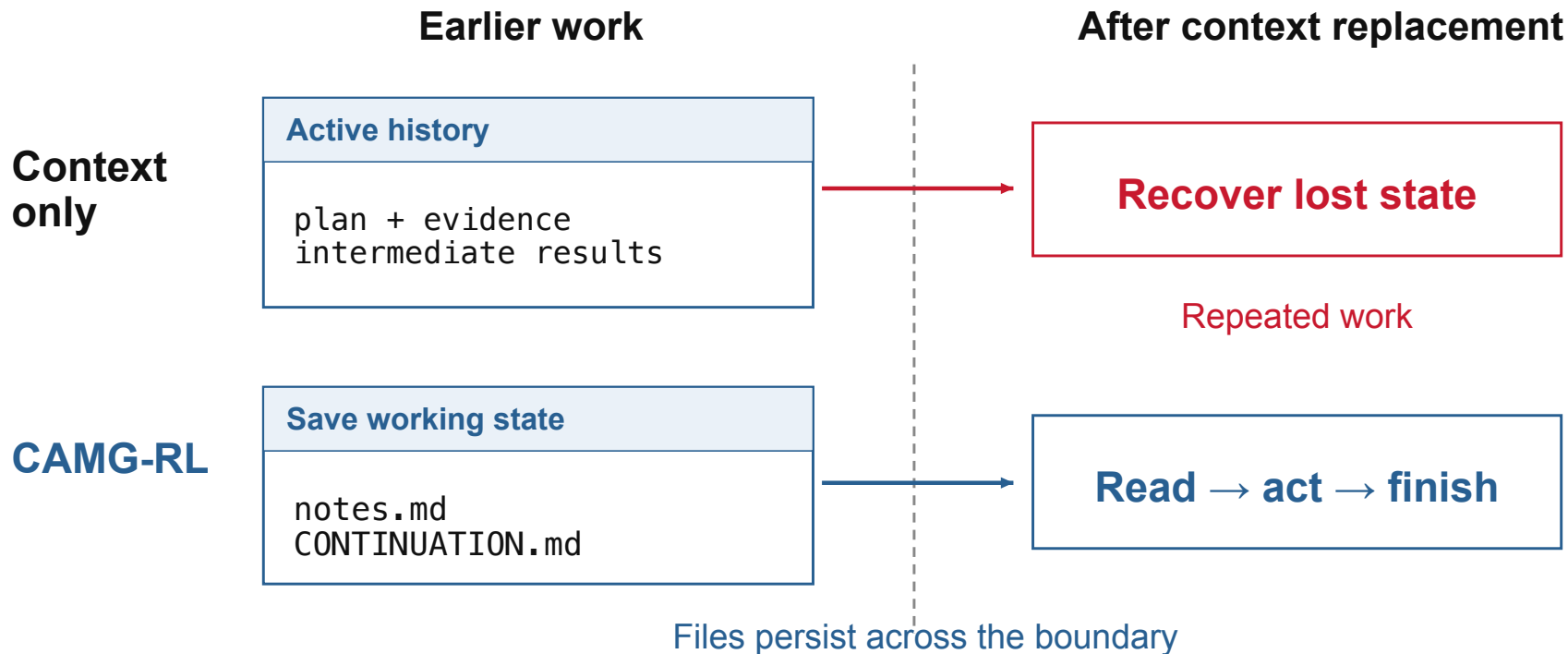
JD.com

Project: [liruiluo.github.io/agentmemorygym](https://github.com/liruiluo/agentmemorygym)

Paper: [arXiv:2609.34422](https://arxiv.org/abs/2609.34422)

Files as learned memory: 54.4% success across four task worlds.

Long tasks need state that outlives context



Durable state lets the next context continue the same task.

Can task reward unlock the prior in file operations?

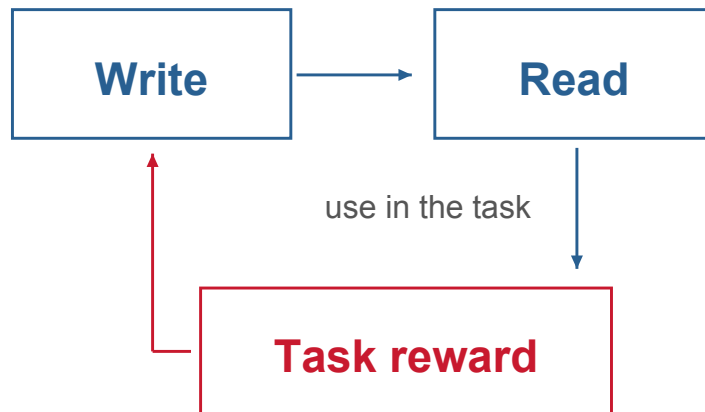
A familiar action language

shell_command

```
cat > notes.md          # store
grep -n fact notes.md   # find
cat notes.md            # read
python revise.py        # revise
```

Available from pre-training

A learned memory strategy



learn when / what to retain

CAMG-RL trains the use of ordinary files from downstream task reward.

CAMG connects four long-horizon task worlds

Shop

Earlier choice



Black case



Later purchase

Match the earlier
color preference

Coding

Repository

```
items()  
→ wrong order  
patch + tests
```

Carry fixes and
test evidence

DeepResearch

Search



Visit sources



Answer

Retain facts and
source evidence

AutoResearch

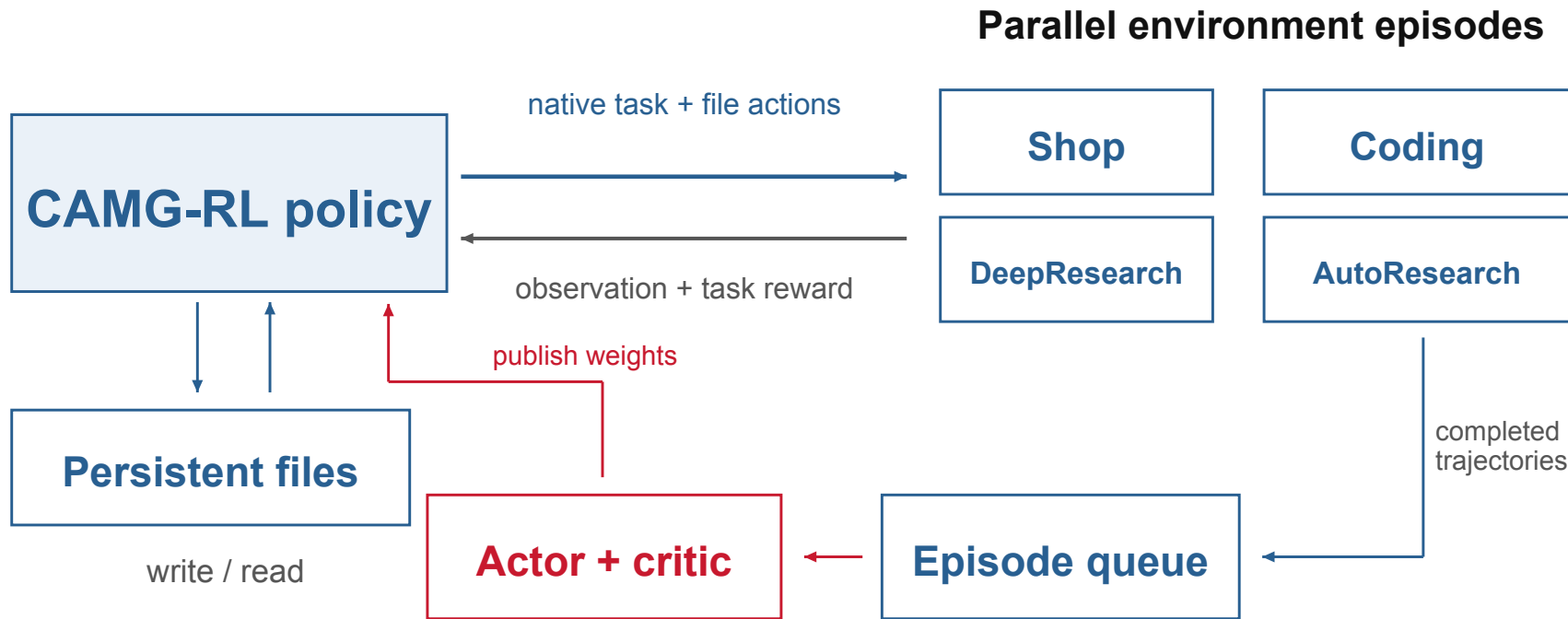
ML workspace

```
train.csv  
model.py  
submission.csv
```

Reuse experiments
and intermediate files

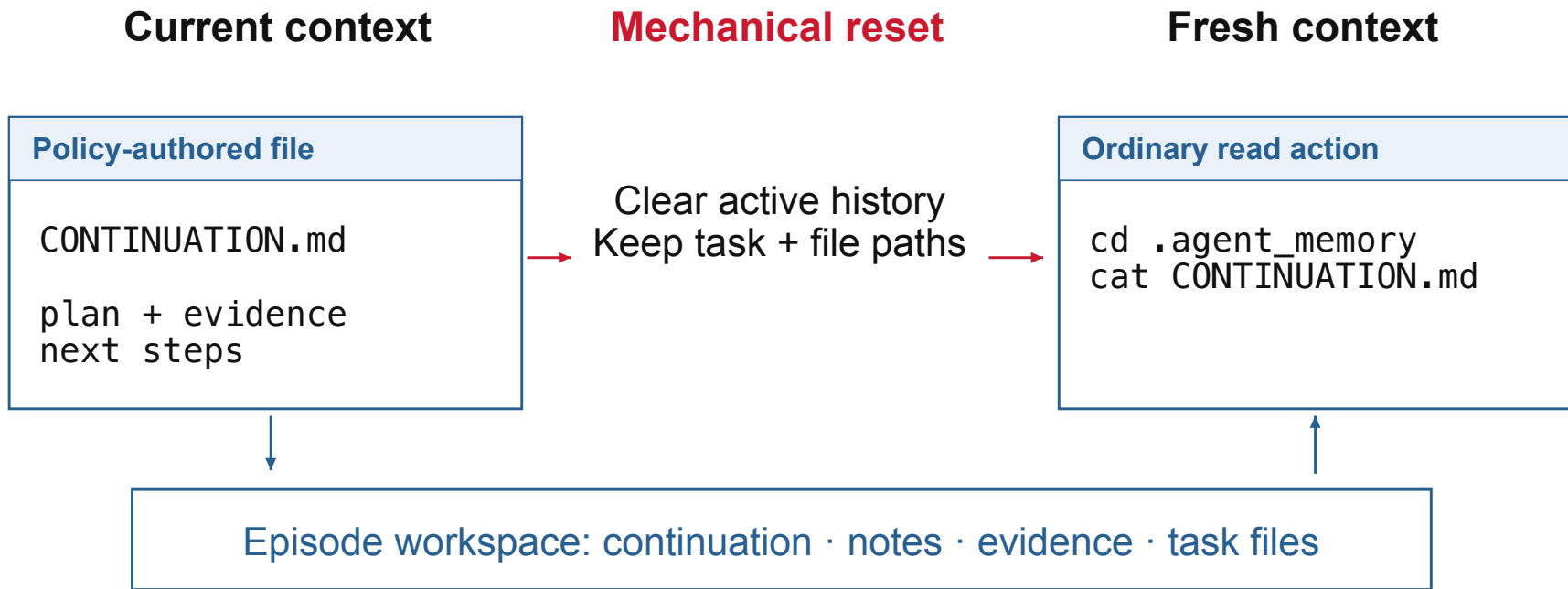
Native tasks + shell access + episode-persistent workspace

One policy learns across all four environments



Fully asynchronous PPO overlaps rollouts and learning.

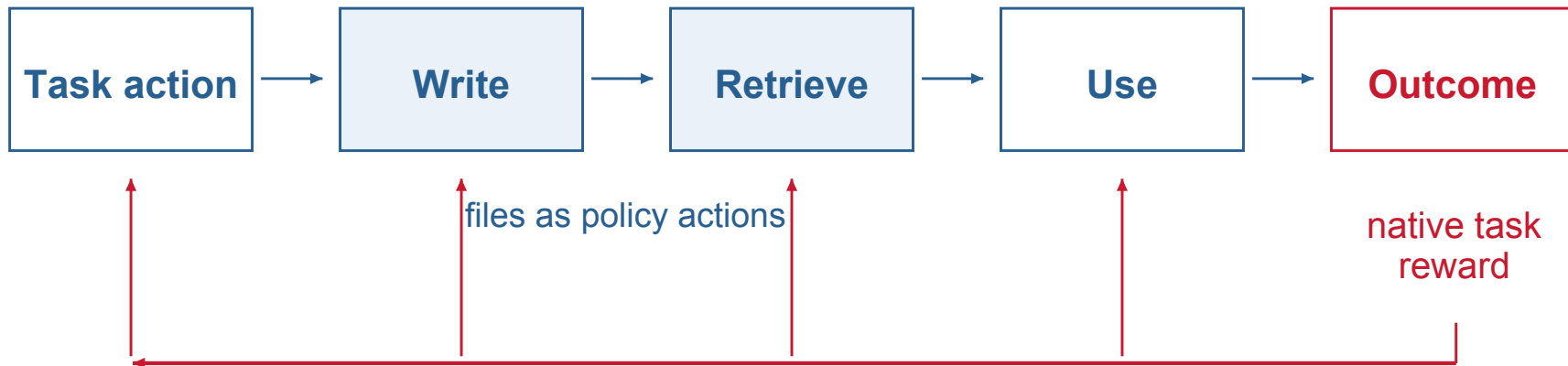
Context changes; the working files remain



The policy writes and reads. The reset contributes no policy action.

Task reward trains the entire memory-use chain

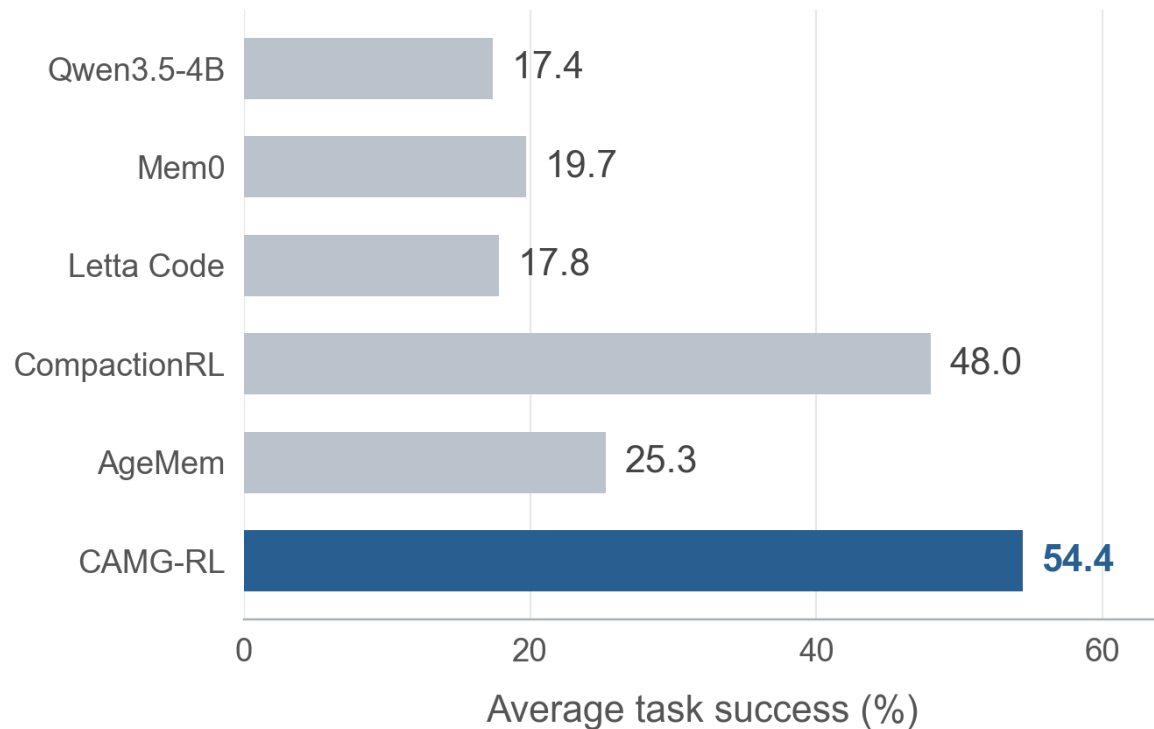
One sampled action trajectory



GAE returns → token-level PPO

Task and memory responses share downstream credit.

CAMG-RL reaches 54.4% average task success



54.4%

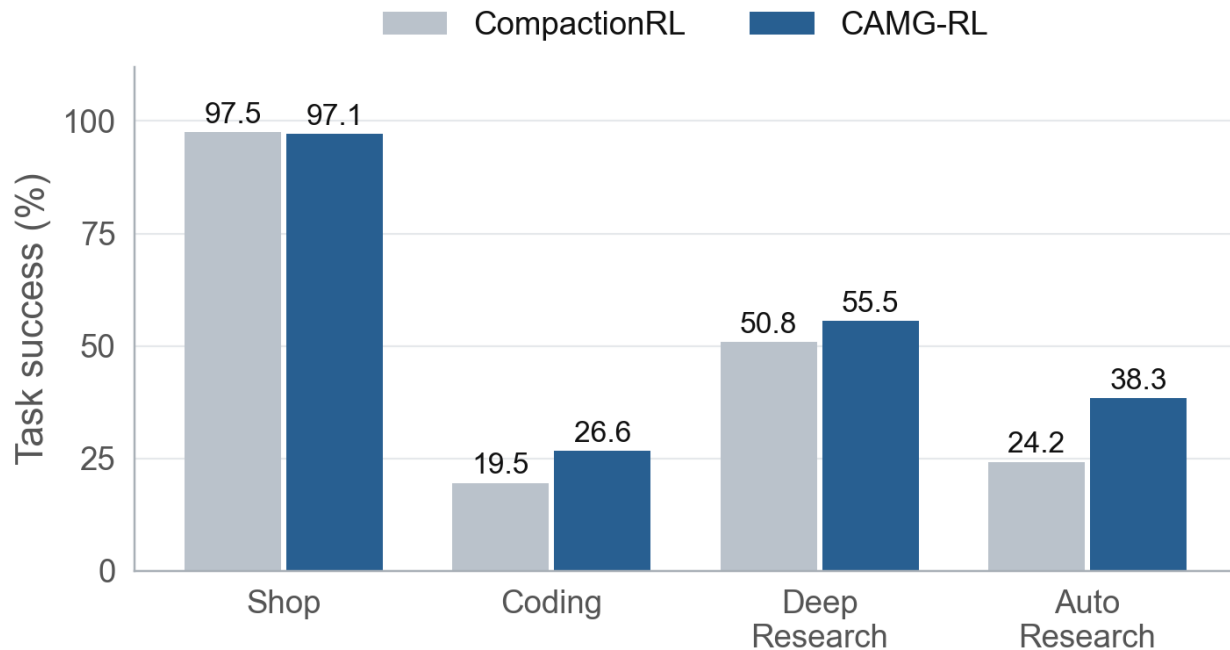
four-environment average

+6.4 pp

over CompactionRL

Learning to use files outperforms the strongest memory baseline.

The largest gains are in Coding and AutoResearch



+7.1 pp

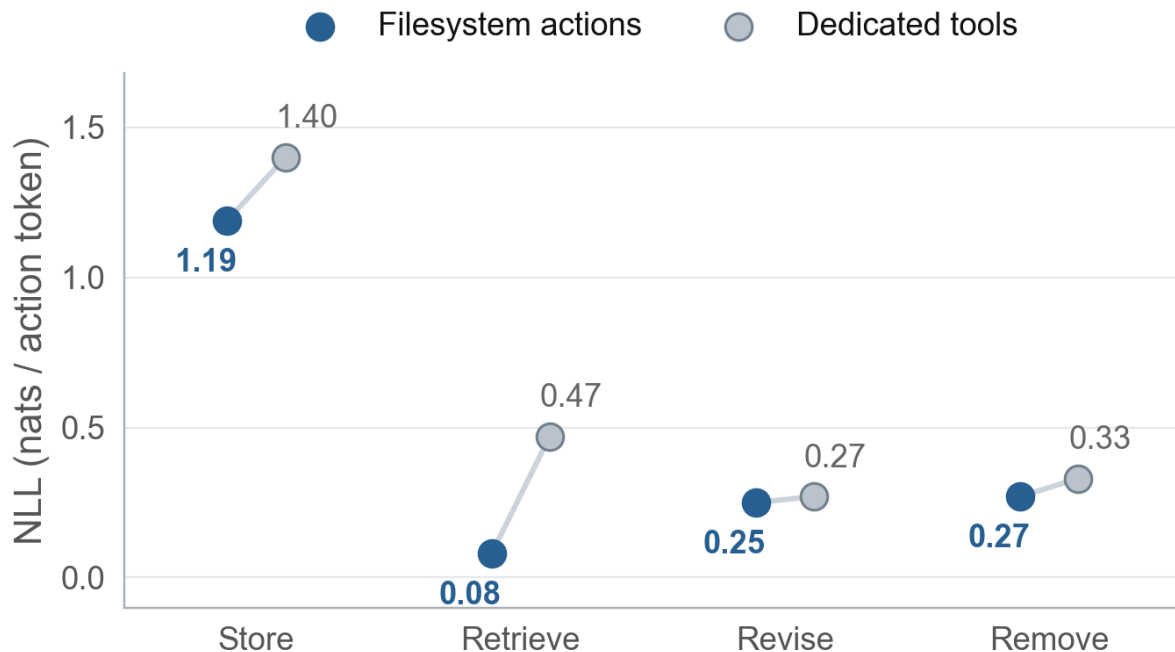
Coding

+14.1 pp

AutoResearch

Shop is near saturation; **Coding and AutoResearch** open the gap.

The frozen base already favors filesystem actions



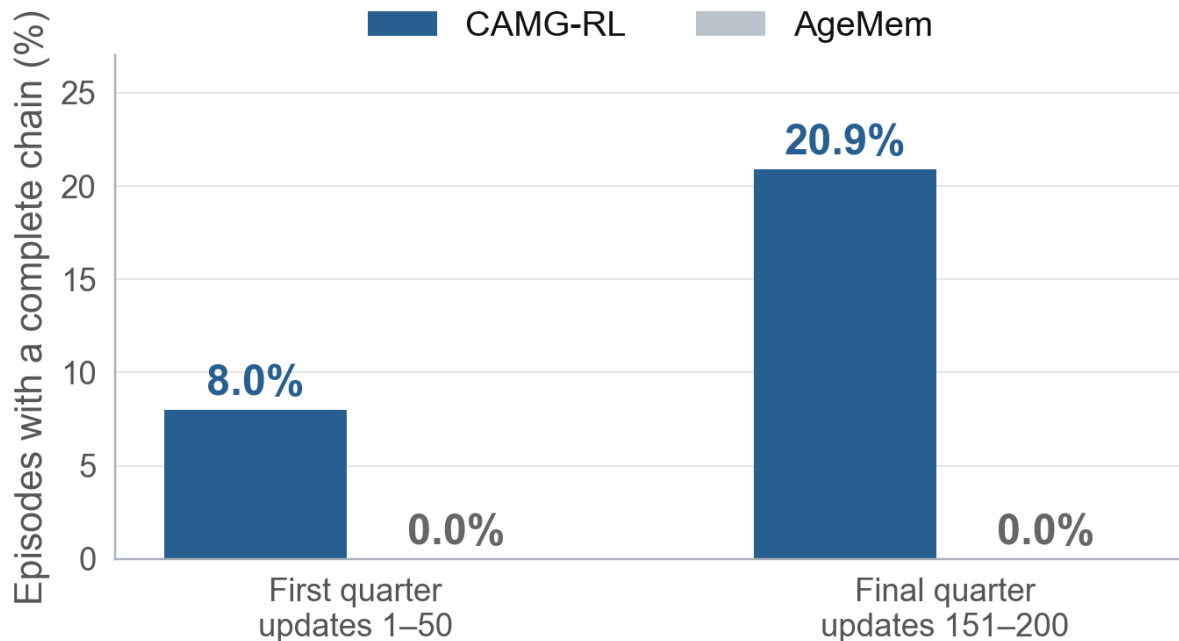
≤ 0.013

nats/token change
when the shell tool is renamed

**The prior extends
beyond the tool
name.**

Familiar action rendering gives reinforcement learning a usable starting point.

RL turns familiar actions into complete memory chains



8.0 → 20.9%

episodes with a complete chain

Store → retrieve → use

Beyond the mandatory
continuation write

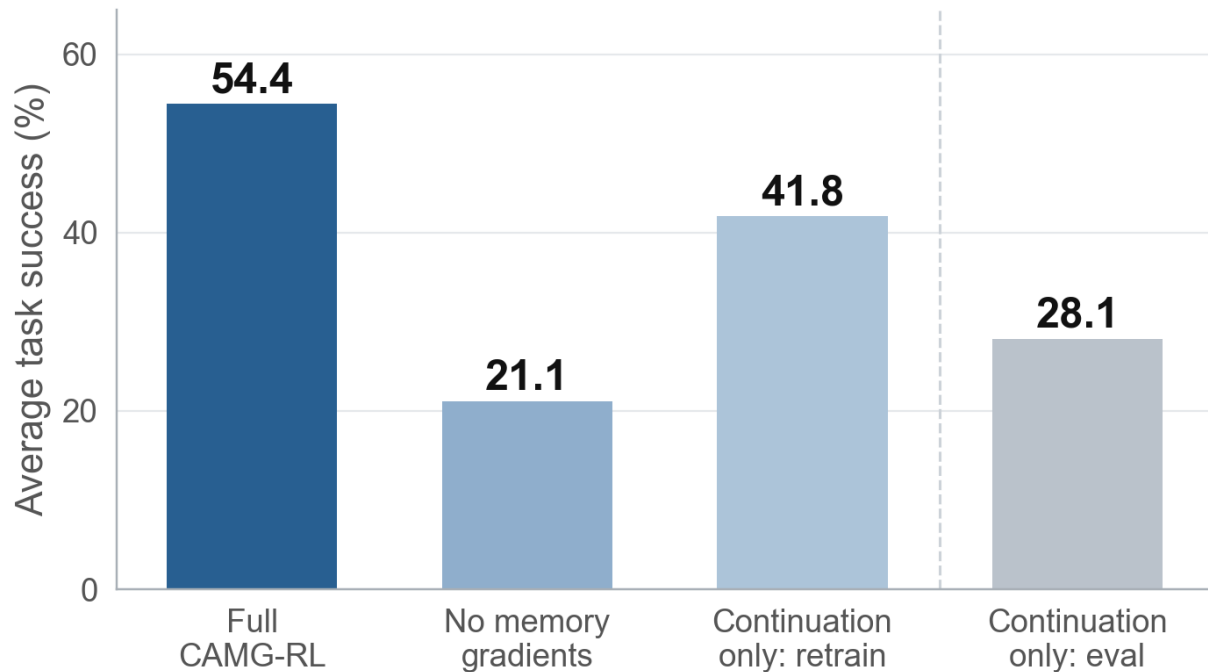
Voluntary memory use grows while AgeMem completes no chain.

Trained competence transfers to external benchmarks

Success rate (%)	SWE-bench Verified	MLE-bench Lite
Qwen3.5-4B	7.6	0.0
CAMG-RL-4B	15.8	4.5
Qwen3.5-35B-A3B	15.6	4.5
Qwen3.5-9B	14.8	4.5
CAMG-RL-9B	27.6	9.1
Qwen3.5-122B-A10B	22.0	9.1

4B and 9B policies are competitive with much larger frozen models.

Memory-action credit and general files both matter



-33.3 pp

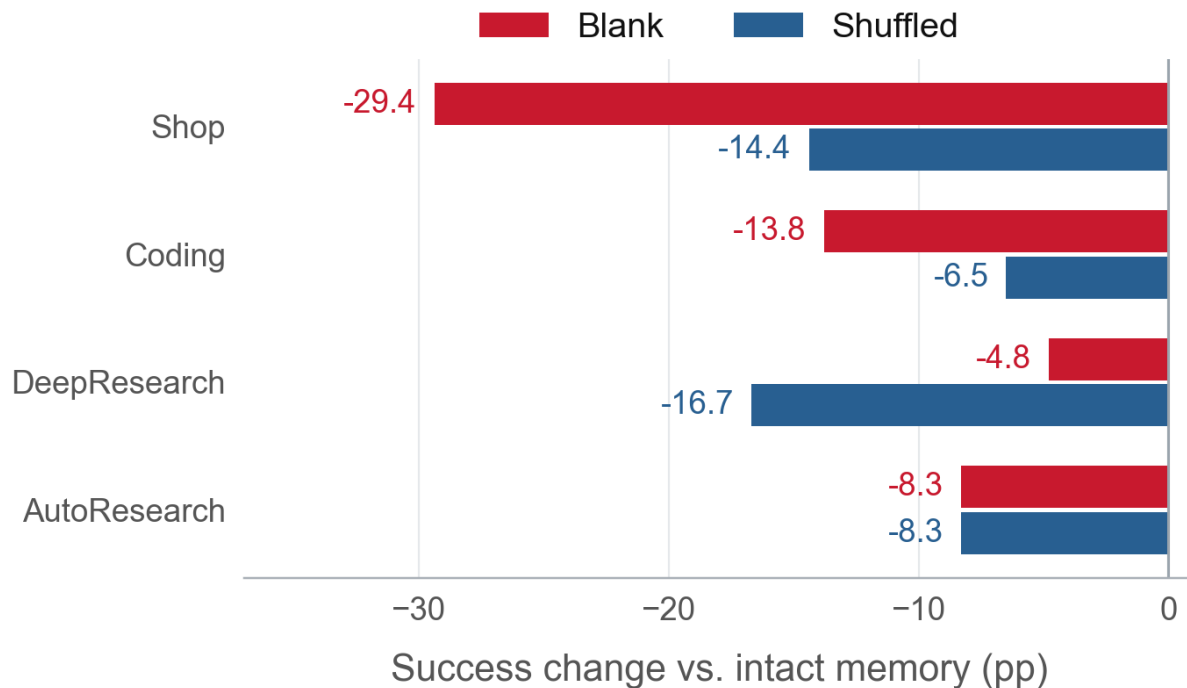
without memory-action
gradients

-12.6 pp

retrained with
continuation-only memory

Learning the file actions and using general memory files both help.

Changing saved content reduces subsequent success



-14.1 pp

blank memory
equal-environment average

-11.5 pp

task-mismatched memory
equal-environment average

Saved content affects later decisions at the same post-reset state.

A coding episode shows the write–read–use chain

Scrapy: repair broken response logging

Before reset · round 18

```
objective: fix response logging
evidence: added _mqpush
         in scrapy/core/scheduler.py
tested: no _mqpush error
next: run tests, confirm
      logging, then submit
```

`.agent_memory/CONTINUATION.md`

After reset · rounds 19–30

```
$ cat \
    .agent_memory/CONTINUATION.md
$ python test_spider.py
$ python -m pytest \
    tests/test_crawl.py -v
```

Hidden tests passed

Return 1.00 · submitted at step 30 / 40

The saved fix and verification plan guide work in the fresh context.

Takeaways

- **CAMG** makes long-horizon memory testable across four native task worlds.
- **CAMG-RL** learns to write, retrieve and reuse files from downstream task reward.
- **Familiar file operations** support learned memory chains and stronger task performance.

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